

Test-1.

1.) A firewall is used to control & secure Network traffic between trusted and untrusted network.

- Block unauthorized traffic
- Allow only required traffic
- Perform NAT
- Monitor & log traffic.

2) Hardware firewall

- dedicated physical appliance

- Higher performance for network security

eg: palo alto,
fortigate

Software firewall.

- Runs as software on a computer / server

- Depends on host resources

eg: Windows defender

3.) ACL

- Mainly controls traffic based on parameters such as source IP, destination IP, protocol and port.
- Commonly configured on routers & switches

Firewall:-

- Provides much more comprehensive security
- Can inspect sessions and application
- Can perform NAT, VPN, IPS, URL filtering, antivirus, SSL inspection etc.

4.) WannaCry was a ransomware attack that affected windows system worldwide

High-Level Sequence:-

- Attackers exploited a vulnerability in SMB.
- Vulnerable systems were infected.
- The malware could spread to other vulnerable systems.
- Files were encrypted and a ransom demand.

Security Lesson:-

- Keep operating systems patched.
- Disable unnecessary services / ports.
- Use endpoint protection.
- Monitor suspicious network traffic.

b) Firewall vendors:-

- Palo Alto networks → Checkpoint
- Fortinet → Sophos.
- Cisco

6. Stateless firewall:-

- It examines each packet independently
- The firewall doesn't maintain the complete state of the connection.

Stateful firewall:-

- It maintains a session / state table
- It records the session, when the server sends the return traffic, the firewall can identify it as part of the established session.

7) NGFW - Next generation firewall.

It provides traditional firewall functions plus advanced security capabilities.

8.) ~~UTM~~ UTM - Unified Threat Management.

UTM combines multiple security

functions into one firewall appliance.

9.) In an enterprise, normally, we don't simply purchase a firewall from a retail shop.

process to buy

Business requirement



Firewall sizing



Vendor / authorized partner



Quotation / BOM



PO



Firewall procurement



License + Support



Installation & Configuration.

10.) Don't select a firewall based only on the number of user

First collect:-

- Number of user
- Internet bandwidth
- Number of application
- VPN Users
- Site-to-site VPN requirement
- HA
- Number/type of interface.
- SSL inspection.
- future growth.

11.) The PA-440 is part of the PA-400 Series

physical connectivity are:-

- 8 x 10/100/1000 ethernet ports
- 1 x dedicated management port
- Console connectivity
- USB
- power connection

Different roles

- Layer-3
- Virtual wire
- Layer-2
- HA

12.) I would compare Palo Alto and Fortigate based on the organization's requirements rather than choosing purely based on vendor preference.

13.) The firewall maintains a session table containing active connections.

At 90% utilization:-

→ The firewall is approaching its session capacity.

→ New connection may eventually be affected if the table reaches its maximum.

→ Performance & connection establishment can impact

→ Existing session may ~~continue~~ continue until aging; but behaviour depends on the firewall.

check:

- Current sessions
- Maximum session capacity
- Session utilization
- Session aging
- Traffic pattern
- Top session-generating hosts
- CPU / Memory

14.) Common firewall objects.

Network objects.

- IP Address
- Subnet

- IP range
- FQDN

Service objects:-

- TCP port
- UDP port
- Service group.

Security objects:-

- address group
- Application group.
- Security profiles
- URL Categories
- USB Groups.

15.) DIPP = Dynamic IP and Port.

It is a type of Source NAT.

Multiple internal / private IP Address can share a smaller number of private IP address by using different source port.

The firewall translate both :-

- Source IP
- Source port.

This allows many internal users to access the internet using one public IP.

16) Intra Zone :-

Traffic between interface belonging to the same security zone.

Interzone :-

Traffic between different security zone.

17) For palo Alto CH.

- Configure
- set deviceconfig system type static
- Set deviceconfig system ip-address 192.168.1.1
netmask 255.255.255.0

→ Commit.

18.) To Create ^{network} object in Palo alto

select ↓
objects
↓
Address
↓
Add.

details:

Name: LAN - Network

Type: IP netmask

value: 10.10.10.0 /24

and select ok, do commit for. Save configuration

19.) step 1- physical / interface status.

check:

→ Interface Status

→ Link up / down

→ Speed / duplex

→ Errors.

Step-2 IP configuration.

check.

- Firewall ip interface.
- Subnet mask
- Default rout.

Step-3 Routing

- 0.0.0.0/0.

Step-4 Security policy.

- Source zone
- Destination zone
- Source IP
- Destination IP
- Application
- Service
- Action.

Step-5

check whether private IP are
being translated to the public IP.

→ 10.10.10.0/24



Source NAT / DIPP



public IP.

Step 6 - DNS

Step 7 - Session / Traffic logs

Monitor → Traffic

→ Allow / Deny

→ Source / destination

→ Application

→ NAT

→ Session and reason.

Step 8 - Packet capture.

If configuration looks correct,
we use packet capture to determine

where traffic is failing.

20. IN palo alto, this is typically configured using Source NAT with Dynamic IP and Port.

policies
↓
NAT
↓
Add.
Original packet.

Source Zone: Trust.

Destination Zone: Untrust

Source Add: Internal Subnet

Destination Add: any

Service: Any

Translated packet:

Under Source translation.

typ: Dynamic IP & Port.

Translated Add

public IP / IP pool / Interface Address

commit.